

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673



231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

March 19, 2026

Patrick Miller
Cadillac, City of
1121 Plett Rd.
Cadillac, MI 49601

RE: Trace Project 26A0292
Client Project NPDES Monthly Requirements- January 2026

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAC Accreditation, Trace certifies that these test results meet all requirements of the NELAC Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAC at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

A handwritten signature in black ink that reads "Timothy W. Brewer".

Tim Brewer
Project Manager

Enclosures



TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 26A0292
Client Project ID: NPDES Monthly Requirements- January 2026

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
26A0292-01	Effluent	Wastewater	TL	01/07/26 12:50	01/08/26 11:50

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ANALYTICAL RESULTS

Trace Project ID: 26A0292
Client Project ID: NPDES Monthly Requirements- January 2026

Trace ID: 26A0292-01 Date Collected: 01/07/26 12:50 Matrix: Wastewater
Sample ID: Effluent Date Received: 01/08/26 11:50

PARAMETERS	RESULTS UNITS	MDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	PQL
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VOLATILE ORGANIC COMPOUNDS BY GC-MS

Analysis Method: EPA 624.1

Batch: T178223

Bromomethane	<0.64 ug/L	0.64	1	01/14/26	nw	01/14/26	nw		5.0
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Surrogates:

1,2-Dichloroethane-d4	107 %	68-133	1	01/14/26	nw	01/14/26	nw		
Toluene-d8	91 %	75-120	1	01/14/26	nw	01/14/26	nw		
4-Bromofluorobenzene	85 %	69-119	1	01/14/26	nw	01/14/26	nw		
1,2-Dichlorobenzene-d4	99 %	72-127	1	01/14/26	nw	01/14/26	nw		

SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS

Analysis Method: EPA 625.1 SIM

Batch: T178137

Benzidine	<0.099 ug/L	0.099	1	01/13/26	als	01/15/26	jlh		0.10
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Surrogates:

Nitrobenzene-d5	90 %	38-130	1	01/13/26	als	01/15/26	jlh		
2-Fluorobiphenyl	72 %	39-118	1	01/13/26	als	01/15/26	jlh	N	
Terphenyl-d14	30 %	14-96	1	01/13/26	als	01/15/26	jlh		

WET CHEMISTRY

Analysis Method: EPA OIA 1677-09

Batch: T178244

Cyanide (Available)	0.0030 mg/L		1	01/15/26	ch	01/15/26	ch		0.0020
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QUALITY CONTROL RESULTS

Trace Project ID: 26A0292

Client Project ID: NPDES Monthly Requirements- January 2026

QC Batch: T178137

Analysis Description: 625 Benzidines SIM

QC Batch Method: GC/MS No Prep

Analysis Method: EPA 625.1 SIM

METHOD BLANK: T178137-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.099	0.099	
Nitrobenzene-d5 (S)	%	82	38-130	
2-Fluorobiphenyl (S)	%	88	39-118	
Terphenyl-d14 (S)	%	48	14-96	

LABORATORY CONTROL SAMPLE: T178137-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.000	0.756	38	16-104	
Nitrobenzene-d5 (S)	%	1.000	0.876	88	38-130	
2-Fluorobiphenyl (S)	%	1.000	0.787	79	39-118	
Terphenyl-d14 (S)	%	1.000	0.425	42	14-96	

Trace Project ID: 26A0292

Client Project ID: NPDES Monthly Requirements- January 2026

QC Batch: T178223

Analysis Description: TTO Volatiles

QC Batch Method: EPA 624 Purge and Trap

Analysis Method: EPA 624.1

METHOD BLANK: T178223-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Bromomethane	ug/L	<0.64	0.64	
1,2-Dichloroethane-d4 (S)	%	107	68-133	
Toluene-d8 (S)	%	92	75-120	
4-Bromofluorobenzene (S)	%	96	69-119	
1,2-Dichlorobenzene-d4 (S)	%	101	72-127	

LABORATORY CONTROL SAMPLE: T178223-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bromomethane	ug/L	50.00	49.2	98	15-185	
1,2-Dichloroethane-d4 (S)	%	30.00	30.5	102	68-133	
Toluene-d8 (S)	%	30.00	28.3	94	75-120	
4-Bromofluorobenzene (S)	%	30.00	29.0	97	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.00	29.4	98	72-127	

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Trace Project ID: 26A0292

Client Project ID: NPDES Monthly Requirements- January 2026

QC Batch: T178206

Analysis Description: Cyanide, Available

QC Batch Method: EPA OIA 1677-09

Analysis Method: EPA OIA 1677-09

METHOD BLANK: T178206-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Cyanide (Available)	mg/L	<		

LABORATORY CONTROL SAMPLE: T178206-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Cyanide (Available)	mg/L	0.05000	0.0521	104	86-118	

MATRIX SPIKE / MATRIX SPIKE DUPLICATE: T178206-MSD1

Original: 26A0292-01

Parameter	Units	Original Result	Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limit	RPD	Max RPD	Notes
Cyanide (Available)	mg/L	0.00261	0.05000	0.0501	0.0563	95	107	82-130	12	11	MS01

Trace Project ID: 26A0292

Client Project ID: NPDES Monthly Requirements- January 2026

QC Batch: T178244

Analysis Description: Cyanide, Available

QC Batch Method: EPA OIA 1677-09

Analysis Method: EPA OIA 1677-09

METHOD BLANK: T178244-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Cyanide (Available)	mg/L	<		

LABORATORY CONTROL SAMPLE: T178244-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Cyanide (Available)	mg/L	0.05000	0.0509	102	86-118	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
MDL	Method Detection Limit
PQL	Practical Quantitation Limit
TIC	Tentatively Identified Compound
<	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds Surrogate control limits
N	The result for this analyte is not covered under our NJ NELAP Annual Certified Parameter List
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.

Definitions and Notes

Trace ID: T178206-MS1

Analysis: EPA OIA 1677-09

Cyanide (Available)	Note MS01 : One of the following: MS, MSD or RPD was out of control. Because two of the results are in control, no data requires qualification.
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Trace ID: T178206-MSD1

Analysis: EPA OIA 1677-09

Cyanide (Available)	Note MS01 : One of the following: MS, MSD or RPD was out of control. Because two of the results are in control, no data requires qualification.
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Page 7 of 8

26A0292

Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 1/8/26	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.6°C)	IR-14 (CF: -1.3°C)	SR1 (CF: -0.2°C)	SR2 (CF: -0.3°C)	Temp Blank	Client Sample
Time: 1502										
Initials: JMC										
Package Description: Cooler										
Package Temp °C	1.0	-0.7								
Representative Sample Temp °C	1.6	1.3								

Sample Receipt

Yes No
☒ ☐ Received on ice or other coolant
☒ ☐ Ice still present upon receipt
☐ ☒ Custody seals present
☒ Trace Courier ☐ Client Drop-off
☐ Yes ☐ No Custody seals intact (if applicable)
☐ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes No N/A
☒ ☐ ☐ All sample containers arrived unbroken and labeled
☒ ☐ ☐ Sufficient sample to run requested analyses
☒ ☐ ☐ Correct chemical preservative added to samples
☐ ☐ ☒ Samples preserved at Trace (if yes, fill in preservation information below)
Preservation - Reagent: _____ LIMS ID: _____ Date/Time: _____
☐ ☐ ☒ Chemical preservation verified, check EMD pH test strip used (if applicable)
☐ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other
☐ ☐ ☒ Air bubbles absent from VOAs (no bubble greater than 6mm) _____

Chain of Custody (COC)

Yes No
☒ ☐ All bottle labels agree with COC
☒ ☐ COC filled out properly
☒ ☐ COC signed by client

Notes:

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